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MANIPAL SCHOOL OF COMMERCE AND ECONOMICS

MINOR PROJECT REPORT

On

Profit-Driven Churn Prediction in Liberalised Utility Market: A comparative Study of Interpretable Machine Learning Techniques

Subject Code- MSBA 5208: Minor Project I

Submitted by

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Under the Guidance of

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DECLARATION

I, **Anurag Dash**, a student of Manipal Academy of Higher Education, Registration No. **251260260051**, hereby declare that the minor project titled **“Profit-Driven Churn Prediction in Liberalised Utility Market: A comparative Study of Interpretable Machine Learning Techniques”** has been prepared by me under the supervision and guidance of **Dr. Rakshit Bhandary**, Assistant Professor, Manipal School of Commerce and Economics, MAHE.

I further declare that this project report is an original work carried out by me and has not been submitted previously for the award of any degree, diploma, fellowship, or any other similar title or prize. The work has also not been published in any journal or magazine.

Place: Manipal

Date: 18 May 2026

Anurag Dash

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Manipal School of Commerce and Economics

Manipal Academy of Higher Education (MAHE)

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Abstract:

Owing to narrow margins and high acquisition costs, in liberalised utility industry, profitability is highly affected by customer churn. Digital industries are impacted by behavioral churn, but utility markets are impacted by different factors like customer price perceptions, contractual conditions, and profits generated from customers. This research has used historical data from a European utility provider having a churn rate of 9.7% with an objective of creating and assessing precision oriented Machine Learning Framework to predict the customer churn. The research pipeline has included cross-validation and hyperparameter optimization to evaluate the supervised machine learning algorithms. Owing to the detrimental effect of severe class imbalance and the operational cost of false positives in customer retention campaigns, model evaluation focuses on precision-recall performance. With an average precision of 0.249, the XGBoost (eXtreme Gradient Boosting) turned out to be the best performing model. The model focuses on high confidence churn identification at a tuned decision threshold by achieving approximately 40 % precision, which states that two out of five flagged customers represent the genuine risk of churn. SHAP (SHapley Additive exPlanations) analysis was performed for robust interpretability. Profitability related variables especially margin_gross_pow_ele and margin_net_pow_ele as well as anticipated contractual and pricing variables such as forecast_meter_rent_12m, are the dominant contributors to risk of customer churn which remain consistent with results. These findings show that churn prediction in utility markets is an economic precision-allocation problem. This aim of the research is to reflect the reliability of precision over accuracy from business specific evaluation explained by the predictive modelling.

Introduction:

Industry Context:

Market liberalization and deregulation have led to a structural transformation in the utility industry. The European and North American electricity and gas market, which were once dominated by vertically integrated monopolies with regulated tariffs and limited choice for consumers, now are converging into competitive retail markets. The European Union's third energy package (2009), for example, required open access to networks, unbundling and better consumer protection, indices for a restructuring of the utility, customer relationship (Fumagalli et al., 2007). Several U. S. states also exercised a retail choice programs of similar nature to enhance market competition.

Consumers have been given a choice through liberalisation to change supplier, compare tariffs and optimize the purchasing strategy for energy (Hartmann et al., 2015). This liberalisation has enhanced competition in the market but mandated operational issues; the concurrent churn of customers for the two energy providers. The fiscal markets induce the introduction of price shocks, regulatory interventions and macroeconomic shocks in the completely deregulated markets have churn become a key output variable with high volatility.

Business Challenge of Churn:

In the utility industry, customer churn presents a very tangible financial impact. Marketing costs, onboarding programs, and regulatory compliance costs make marketing costs and profit margins structurally narrow, making acquiring a new customer' many more times expensive than maintaining an already established customer, as research from before shows (Neslin et al., 2006; Lemmens& Gupta, 2020). Utility churn is often an economic calculus decision, especially in telecom or digital ecosystems that are predominantly activity driven by behavioral signals. As customers near contract renewal periods, they compare through time projections or prices and tariffs and foresee consumption trends (Hartmann et al., 2015). Consumer behavior also is exogenously determined by weather and seasonality, which convey high dimension and nonlinear data structures.

However, the key to point is that only some proportion of customers defect in a period allowing the accumulation of extensive historical information and therefore making churn somewhat rare events. This harsh imbalance raises extraordinary difficulties in predictive modelling approaches (Burezand Van den Poel, 2009; He and Garcia, 2009).

Research Gap: Precision as a Strategic Imperative

Even though large amount of work has been done on predicting customer attrition, the current techniques have their own drawbacks, which are more suitable for the liberalized utility markets. While previous efforts like Logistic Regression provide a model that is easy to interpret, it fails to incorporate nonlinear interactions between costs, price and customer consumption patterns (Cramer, 2002; Huang et al., 2012). The use of Kernel based methods such as Support Vector Machines suffers from degeneracy and instability as they are very sensitive to choosing the optimal hyperparameters and class imbalances (Coussement & Van den Poel, 2008).

In tabular representation, advanced ensemble methods such as Random Forest, Gradient Boosting among others have proven very effective to gain predictive power (Breiman, 2001; Friedman, 2001; Chen & Guestrin, 2016). However, most of the literature focuses on optimizing accuracy or F1score, without calibrating the evaluation measure to the structure of economic costs and benefits of retention choices (Verbeke et al., 2012; Verbraken et al., 2013). In the utility markets false positives are especially expensive, as they lead to false incentives (damaging thin margins) by detecting a non churner as a churner. No single classification measure captures this asymmetry from the decision theory point of view.

There is clearly an interesting, well versed, and under or untapped area of research here. The biggest challenge for utility churn prediction lies in finding a modelling approach that both achieves good overall accuracy (which, due to class imbalance, necessarily balances recall and precision) and aligns well with the bottom line. Indeed, precision is an absolutely critical measure for very cost sensitive situations such as this, in that we want to ensure we truly do identify customers at danger of churning (Saito & Rehmsmeier, 2015). Previous utility studies dealt with either the usual consumption prediction or the across the board class predictions, but without the precision oriented insights (Popescu et al., 2021; Alghamdi et al., 2022).

Proposed Approach: A Precision-Recall Balanced Framework:

This project is intended to be an ambitious exercise to develop a reliable efficient machine learning workflow for improving customer churn prediction. Our goal is to apply a rigorous retrospective analysis to develop and validate our models on a high class imbalance problem simulated by a historical database of a European energy provider.

Applying a rigorous assessment framework, we benchmark 4 algorithms Logistic Regression, Support Vector Machine, Random Forest, and XGBoost consistent with best practices within comparative modelling literature (Lessmann et al., 2015). The procedure uses Bayesian hyperparameter optimization with stratified 5fold cross validation, optimizing for fine balance of precision and recall in F1score, with selection fitting explicitly featuring precision as the key criterion of choice.

Additionally, our method can be enhanced by using SHAP (SHapley Additive exPlanation) models, which provide interpretability of the models in terms of defining economically meaningful drivers (Lundberg & Lee, 2017; Guidotti et al., 2018). This is analogous to the current call for transparency and auditability in decision systems with high–financial stakes (Rudin, 2019). On top of the empirical validation, we apply Statistical Validations, where we use Mc Nemar's test to prove that the performance difference we observed empirically are statistically sound and repeatable (Dietterich, 1998).

Contributions:

This research makes three primary contributions to the literature on utility-sector churn prediction:

1. **Precision – Recall Balanced Model Selection**
   In consistent with previous findings on supremacy of gradient boosting in structured data (Chen & Guestrin, 2016), we show that under severe class imbalance, XGBoost performs much better than linear, kernel-based, and bagging-based alternatives by achieving the strongest balance between precision and recall.
2. **Economic Interpretability of Churn Drivers**
   We experimentally show that churn is majorly driven by economic factors like customer profitability and anticipated future costs through SHAP analysis which support the idea that churn is a rational, forward-looking economic decision (Verbeke et al., 2012; Lemmens & Gupta, 2020).
3. **Deployable and Auditable Framework for Utilities**
   In order to align predictive modelling with real-world decision-making constraints, we provide a completely reproducible workflow which integrates Bayesian optimization, cost-sensitive learning, statistical validation, and explainable machine learning (Elkan, 2001; Bahnsen et al., 2015).

Literature Review

In data-driven industries, customer retention is particularly more cost-effective than acquiring them due to the longstanding concern of customer churn prediction. In liberalized utility markets, the existing literature establishes a strong theoretical and foundation basis for comprehending customer churn by identifying key economic and relational drivers and the modelling strategies used to capture them. Most researchers agree that churn is not a random occurrence rather a calculated customer decision which is driven by economic utility, behavioural cues and contractual constraints.

1. Price Sensitivity and Perceived Value

In liberalized utility markets, where switching barriers are minimal, price sensitivity remains as one of the key drivers of customer churn (Hartmann et al., 2015). Based on Transaction Utility Theory (Thaler, 1985), negative perceived value trigger churn and customers assess tariffs relative to internal or market-based reference prices. Due to such inherently forward-looking economic evaluation, customers respond very strongly to expected losses than to past expenditures. In this scenario, for capturing expected future costs rather than previous billing patterns, forecast-based pricing variables like *forecast_meter_rent_12m* act as critical predictive signals. The importance of pricing signals in influencing switching behaviour is supported by the experimental results of earlier research in utility sector (Popescu et al., 2021; Alghamdi et al., 2022). The importance of capturing non-linear price sensitivity using advanced machine learning algorithms from a modelling standpoint has been highlighted as per earlier research (Ahmad et al., 2019; Caigny et al., 2020). Gradient Boosting, an ensemble method (Friedman, 2001; Chen & Guestrin, 2016) implicitly model relative positioning of price without requiring explicit feature engineering, making them highly effective in learning non-linear relationships.

2. Customer Profitability and Firm-Customer Alignment

Customer profitability is duality in nature. It plays as both an outcome metric and a behavioural indicator of churn propensity. Lower-margin customers are particularly more sensitive towards price and more responsive towards competitors offer and they align significantly low with firm-level retention strategies as per the customer lifetime value (CLV) theory (Lemmens & Gupta, 2020; Verbeke et al., 2012). Characterized by narrow margins and heavy competition, this relationship is crucial in energy markets. Backed by empirical studies, profit driven churn modelling highly improves decision-making as compared to purely accuracy-driven methods (Verbeke et al., 2012; Verbraken et al., 2013). In our study, *margin_net_pow_ele* and *margin_gross_pow_ele,* which are profitability related strong churn predictors reflect hidden economic dissatisfaction. Earlier research frequently integrates interaction terms or profit-based segmentation to capture these dynamics (Burez & Van den Poel, 2009; Glady et al., 2009) from a methodological perspective. However, such interactions can be automatically captured through recursive partitioning by ensemble methods like Random Forests (Breiman, 2001) and boosting models (Friedman, 2001) by providing a flexible and scalable alternative to manual feature engineering.

3. Contractual Status and Relationship Depth

Churn behaviour is highly driven by relational factors like customer tenure and contractual commitment. Stronger relationships between customers and firms reduce churn by increasing both psychological and economic switching barriers (Neslin et al., 2006; Fumagalli et al., 2007) according to theoretical viewpoints based on relationship marketing and switching cost theory. These relational dimensions are captured by predictors like *num_years_antig* (tenure), *nb_prod_act* (number of active products), and contract renewal indicators. Experimental results indicate that when switching costs are temporarily reduced, churn risk increases contract renewal periods (Hartmann et al., 2015). Relational variables frequently show lower predictive performance in competitive markets (Popescu et al., 2021) as compared to economic variables. Earlier studies have highlighted the role of temporal feature engineering and survival analysis techniques in covering the contract related dynamics (Glady et al., 2009) from a modelling standpoint. Contrary to this, XGBoost, a tree-based ensemble algorithm can directly learn non-linear tenure effects and renewal patterns without needing temporal transformations.

4. Consumption Patterns and Usage Shocks

By capturing underlying shift in customer activity, consumption behaviour provides additional understanding in churn dynamics. Consumption patterns are valuable early churn indicators (Huang et al., 2012; Idris et al., 2012) as indicated by behavioural theories that changes in observed actions frequently precede decision-making. Features like *cons_12m*, *cons_last_month*, and *forecast_cons_12m* capture both anticipated future demand and historical usage. Even if absolute consumption levels are frequently weak features, variation from expected patterns like increased volatility or sudden fall can signal structural changes like relocation or reduced activity (Chico, 2012). When combined economic variables with consumption dynamics, it improves churn prediction as confirmed by experimental studies (Ahmad et al., 2019; Popescu et al., 2021). Derivative variables like percentage change and consumption volatility are commonly used to increase predictive performance (He & Garcia, 2009). To capture non-linear deviations without imposing strict parametric assumptions, ensemble methods and particularly boosting algorithms are effective.

5. Customer Profile and Segmentation

Customer segmentation helps to explain differences in churn behaviour within a wider context. According to segmentation theory, groups of customers differ in their price sensitivity, loyalty, and switching tendency based on their channel of acquisition and activity with the business (Neslin et al., 2006). So there for those variables like channel_sales and activity_new are moderating variables not drivers of churn. The customers obtained via price-driving channels will have a higher churn rate due to low switching costs and low brand attachment (Hartmann et al., 2015). According to empirical evidence, although these categorical variables are predictive, they are not usually stronger than economic variables (Alghamdi et al., 2022). When it comes to methodology, it's very important to effectively handle categorical variables. Target encoding and tree-based methodologies are useful in this regard (Lessmann et al. 2015, Caigny et al. 2020). XGBoost and other frameworks that implement gradient boosting improve this ability by capturing subtle segment-level differences through learning.

The model selection is directly influenced by the complex and non-linear relationships among contractual status, price perception, profitability and consumption behaviour. Statistical approaches such as logistic regression and discriminant analysis (Cramer, 1975) were previously followed to study the churn behaviour. Although interpretable, the linear assumption of these models makes them struggle to capture heterogeneous interactions between economic and relational drivers. The development of ensemble techniques severely advanced churn modelling. Random Forests (Breiman, 2001) efficiently model complex interactions without overfitting because of the robustness through bagging and variance reduction. Gradient Boosting Machines (Friedman, 2001) perform much better on imbalanced dataset by sequentially correcting prediction errors. The trade-off between precision and recall and performance under severe class imbalance was because of the methodological disagreement between bagging and boosting. False positives are expensive in retention settings because the customers who wouldn't have churned might have gotten incentives. Hence, precision becomes more crucial than accuracy. For structured tabular data, advanced boosting techniques like XGBoost (Chen & Guestrin, 2016) are more suitable because of the methods like regularization, parallelization, and optimized tree building. Boosting techniques when optimized properly, outperform linear baselines and exceed bagging techniques marginally across industries including telecommunications (Vafeiadis et al., 2015; Idris et al., 2019).

Irrespective of extensive study in telecommunications and finance, churn in liberalized energy markets remains considerably less explored, which reveals a critical gap in research. Generally, utility industry operates under constraints of regulation, and for which churn rate also remains low. Mostly, churn in this industry is a response to expected meter-related charges (forecast_meter_rent_12m), tariff structures, and profitability alignment (margin variables).

Existing machine learning applications in utilities primarily focus on load forecasting (Popescu et al., 2021) or demand classification (Alghamdi et al., 2022), rather than economically driven customer defection. Moreover, many studies emphasize accuracy or F1-score optimization without aligning evaluation metrics with business cost structures.

This reveals two critical gaps:

1. **Limited integration of interpretable machine learning with profitability-centered churn modelling.**
2. **Insufficient emphasis on precision-oriented evaluation tailored to retention economics.**

Our work expressly frames the model as a strategic choice as opposed to statistical one to address the gap. Under severe class imbalances, structured ensemble algorithms like Random Forest which is a bagging technique and Gradient Boosting variants are performed. As per the experimental results, XGBoost provides higher precision-recall as compared to alternate ensemble techniques in this utility churn context. As per the SHAP test, profitability features like margin_net_pow_ele, margin_gross_pow_ele and expected economic charges (forecast_meter_rent_12m) are the key features driving churn, confirming the economic utility hypothesis. The research is further enhanced by Bayesian hyperparameter tuning, explanation via SHAP and statistical significance testing which encapsulates the model fitting according to the retention strategy for the given profitability constraint. In deregulated utility industry, precision becomes more significant over accuracy where the tradeoff between narrower margins and more expensive retention inducements still stands.

To formulate and validate a precision focused, economically resourced cost effective churn prediction framework for deregulated utilities; translating the ML findings into economic decisions under class imbalance.

RQ1: How should utility customer churn be defined as an ordinary classification problem or as an economic decision problem with asymmetric costs?

RQ2: How do the various machine learning categories (linear, kernel, bagging and boosting) successfully encode the nonlinear and interaction effects endemic to utility data churn narratives?

RQ3: How can the existing churn prediction models be finetuned to maximize precision at balanced or high levels of recall in the context of stark class imbalance?

RQ4: What are the idiosyncratic economic incentives behind customer disaffection, and are we able to employ explainable AI techniques to reveal competitive, strategic or exogenous factors?

RQ5: Are the performance differences between model variants statistically significant and consistent across validation trials?

Methodology:

Data Description and Preprocessing

The SME client dataset from a European energy provider was specifically selected for this study due to its relevance with high-value B2B churn but also because of its depth and consistency of detailed profitability metrics, which is the core of the study's hypotheses. A single provider dataset was chosen to keep the analysis under control and extract deep features from this dataset. The economic variables under analysis include net_margin and margin_net_pow_ele. This is in line with previous churn studies in the energy sector. These studies show that customer behaviour is focused on profitability (Popescu et al., 2021; Lemmens & Gupta, 2020).

The data was organized in two pieces.

• The Client Snapshot (client): is a static table containing 14,606 records and 26 features, including cons_12m (annual consumption), net_margin (profitability), origin_up (acquisition channel), and a binary churn label, which indicates if the contract was not renewed.

• The price history (price): is a time series table with 193,002 records detailing the daily price for electricity for each customer in 2015–2016.

The target variable for churn prediction was extremely imbalanced with only 9.7% of the clients that actually churned. A scenario that is realistic and widely reported in the literature related to churn prediction (Burez & Van den Poel, 2009; He & Garcia, 2009) that presents a key modelling challenge.

Our preprocessing was driven by domain knowledge and tree-based model needs.

• Temporal Feature passage Date variables were transformed into economically meaningful predictors. The expiration dates of contracts were turned into signals related to renewal; Additionally, features based on the year and duration were created to estimate the switching windows. The transformations were directly linked to switching cost theory as well as renewal-based triggers of churn (Neslin et al., 2006; Hartmann et al., 2015)). We turned the datetime features into predictors. For instance, the date_end created the contract_end_year, which is a clear indicator of the approaching decision point of the customer.

• The average price was not sufficient to engineer the price sensitivity. We constructed features like diff_Dec_mean_price_off_peak_var (the difference between a client's December price and their annual mean) to capture sticker shock a customer may feel before renewal, a hypothesized churn trigger, and an already suspected churn mechanism in the utilities market (Popescu et al., 2021).

• The categorical variables were label encoded (e.g. channel_sales, origin_up) which is entirely tree-compatible for further experiments using XGBoost etc. According to Breiman (2001) and Friedman (2001), tree-based methods, in contrast to linear models, are invariant to monotonic transformations.

Descriptive Statistics

To characterize the dataset central to our analysis, descriptive statistics were computed for the primary numerical features selected a priori based on domain expertise in the utility sector and their hypothesized relationship to customer churn in prior literature [e.g., Lemmens & Gupta, 2020]. This exploratory analysis was conducted solely on model input features and without conditioning on the churn label to prevent target leakage and ensure an unbiased understanding of the data structure.

There are significant variation across some important commercial and operational variables in the data set. Two of the profitability related variables, net_margin and margin_net_pow_ele, are widely scattered, showing variation in tariff design and customer level profitability. Similarly, annual consumption (cons_12m), a proxy for SME size and the maximum demand (pow_max), shows wide variation.

This heterogeneous evidence suggests one can expect interactions effects and nonlinear thresholds (in relation to profitability, prices, and volume dependent variables) to be key. This also further suggests adopting flexible (can incorporate interactions) modelling approaches in particular gradient boosting approaches such as XGBoost (Lessmann et al., 2015).

Table 1: Descriptive Statistics

		count	mean	std	min	median	max	skewness	kurtosis
1	margin_net_pow_ele	14606.0	24.56	20.23	0.0	21.64	374.64	4.47	35.9
2	margin_gross_pow_ele	14606.0	24.57	20.23	0.0	21.64	374.64	4.47	35.89
3	net_margin	14606.0	189.26	311.8	0.0	112.53	24570.65	36.57	2642.97
4	cons_12m	14606.0	159220.29	573465.26	0.0	14115.5	6207104.0	6.0	42.69
5	cons_last_month	14606.0	16090.27	64364.2	0.0	792.5	771203.0	6.39	47.76
6	forecast_cons_12m	14606.0	1868.61	2387.57	0.0	1112.88	82902.83	7.16	147.43
7	forecast_meter_rent_12m	14606.0	63.09	66.17	0.0	18.8	599.31	1.51	4.49
8	pow_max	14606.0	18.14	13.53	3.3	13.86	320.0	5.79	59.2

Correlation Analysis and Implication for model selection

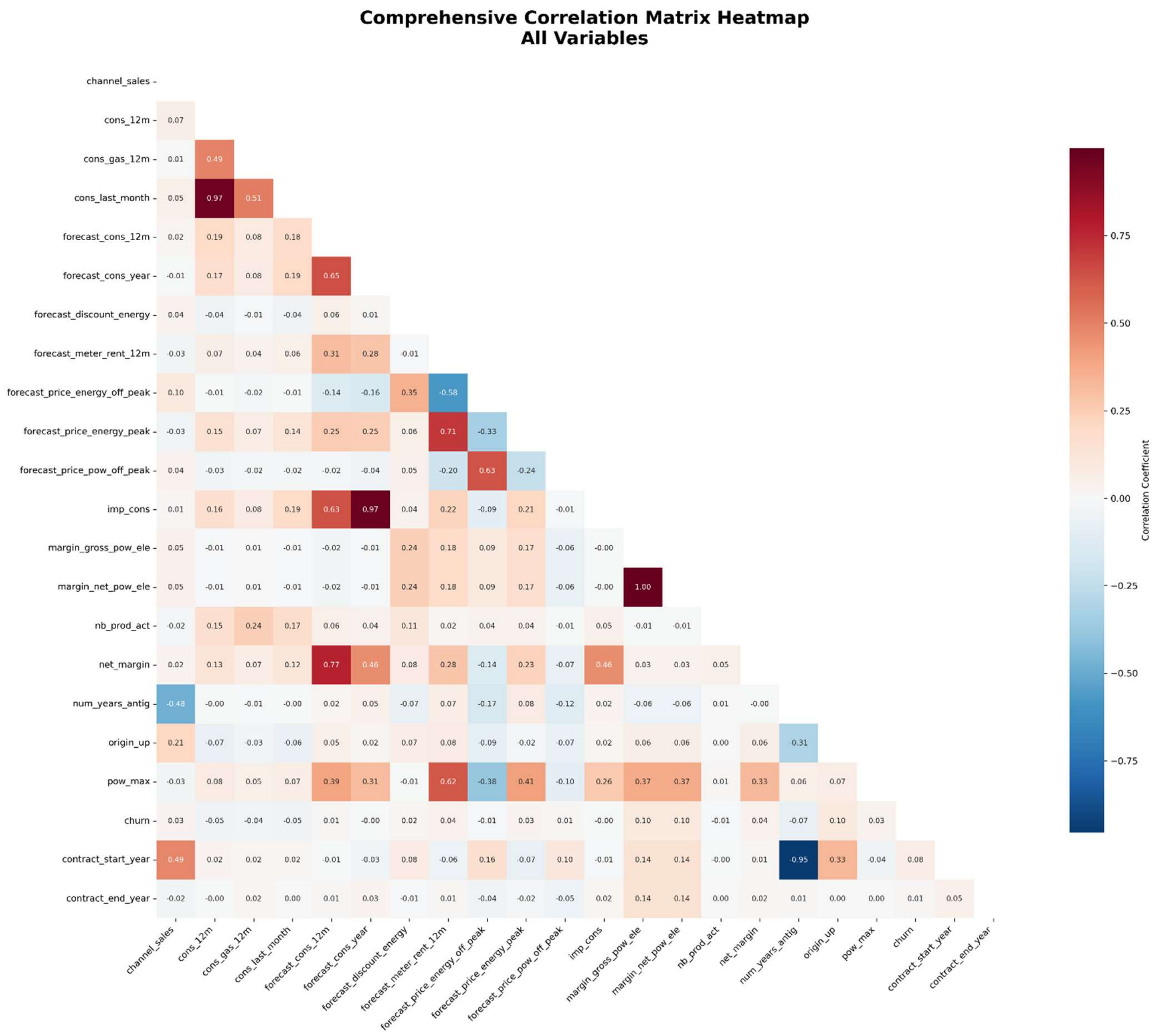

Fig. Correlation Plot

The figure illustrates the pairwise correlation structure of all the engineered variables. Although some consumptionrelated features have internally high correlation (for instance cons_12m and cons_last_month), the profitability and pricing variables have low linear dependence, which indicates that their effective importance in later stages is not mere consequence of collinearity.

A bivariate correlation analysis was performed on the churn indicator and all the other numerical indicators. As recommended by Cohen (2023), correlation coefficients can be considered substantively significant if the absolute value of correlation coefficients are greater than 0.2. No variables being examined exceed this threshold, as shown in Table X. The profitability variables of margin_net_pow_ele (|r| = 0.096), margin_gross_pow_ele *(|r| = 0.096), and origin_up *(|r| = 0.096) had the greatest observed correlation coefficients (in the "small" effect size category).

On the basis of the above insight, it appears that linear relationships are not able to offer explanatory power for customer attrition in this industry, with attrition arguably being a consequence of complex, nonlinear joint relations across a wide range of consumption, contract and economic related variables. This naturally indicates the usage of nonlinear ensemble methodologies such as (very high dimensional) Random Forest and XGBoost, which are able to incorporate third, fourth (up to all higher order) interaction effects i.e. beyond linear correlations (Breiman, 2001; Chen & Guestrin, 2016).

Inferential Statistics

To ensure a rigorous and theoretically grounded comparison, we model customer churn as a probabilistic prediction problem where, for each customer i, the objective is to estimate the conditional probability of churn $\hat{p}_i = P(y_i = 1 \mid X_i)$, with $y_i \in \{0,1\}$. Predictions are converted into decisions using a threshold rule $\hat{y}_i = \mathbb{I}(\hat{p}_i \geq \tau)$, where the threshold $\tau$is later optimized under a profit-maximization framework.

We evaluate four classes of models representing distinct functional assumptions:

(i) Logistic Regression, defined as $\hat{p}_i = \sigma(X_i \beta)$, serves as a linear benchmark under log-odds separability and is estimated via maximum likelihood.

(ii) Support Vector Machines, which solve a margin-maximization problem $\min_{w,b} \frac{1}{2} \| w \|^2$subject to classification constraints, allow non-linear decision boundaries through kernel transformations.

(iii) Random Forest, an ensemble of decision trees $\hat{f}(X) = \frac{1}{B} \sum_{b=1}^{B} T_b\,(X)$, captures higher-order interactions via recursive partitioning and reduces variance through bootstrap aggregation.

(iv) Extreme Gradient Boosting (XGBoost) models predictions as an additive expansion $\hat{y}_i = \sum_{t=1}^{T} f_t\,(X_i)$, optimized using second-order gradient methods with explicit regularization to control model complexity.

This structured portfolio enables the empirical identification of the underlying functional form governing churn behavior—whether linear, non-linear, or interaction-driven. Importantly, model evaluation is not restricted to statistical accuracy but is embedded within a decision-theoretic framework, where optimal predictions are those that maximize expected profit:

$$\max \sum_i [\hat{p}_i \cdot V_i - (1 - \hat{p}_i) \cdot C_i]$$

Empirically, XGBoost outperforms alternative models across precision–recall trade-offs and demonstrates superior stability in minority-class detection under severe class imbalance, making it the preferred model for profit-oriented churn intervention.

Critical Implementation Detail

We have a relatively high (9.7%) churn rate in our data set which is typical of a liberalized utility market. Rather than artificially oversampling data patterns (say synthetic oversampling, SMOTE), which would further fit the empirical distribution, and distort the real financial balance of the models, we solved class imbalance internally via model internal balancing:

- For XGBoost, the scale_pos_weight parameter was tuned to penalize misclassification of churners.
- For other models, class weights were adjusted proportionally to class frequency.

This approach ensured that:

- The data distribution was left unchanged.
- Learning minority patterns was not artificially boosted.
- Business evaluation was realistic.

Table 2: Hyperparameter Search Spaces and Optimal Values

Model	Hyperparameter	Search Space	Optimal Value
XGBoost	n_estimators	Integer(100,400)	236
	max_depth	Integer(3,6)	4
	learning_rate	Real(0.01,0.2, log-uniform)	0.013690677
	subsample	Real(0.6,1.0)	0.822936169
	colsample_bytree	Real(0.6,1.0)	0.777933005
	min_child_weight	Integer(1,10)	3
	gamma	Real(0,5)	4.593612608
	reg_alpha	Real(1e-4,1.0, log-uniform)	0.000418593
	reg_lambda	Real(1.0,10.0)	7.763972782
Random Forest	n_estimators	Integer(50,200)	97
	max_depth	Integer(5,15)	9
	min_samples_split	Integer(2,5)	5
	max_features	Categorical(['sqrt', 'log2'])	log2
SVM	C	Real(1e-2,1e2, log-uniform)	0.436933995
	gamma	Real(1e-3,1e-1, log-uniform)	0.028539837
Logistic Regression	C	Real(1e-2,1e2, log-uniform)	0.010340016
	solver	Categorical(['liblinear'])	liblinear

After running hyperparameter tuning, XGBoost came out on top for churn prediction.

XGBoost (Top Performer): Used not-too-deep trees (max_depth=4) that would allow to learn yes, but not overfit. Had a small learning rate (0.0137), but with sufficient number of estimators (236) would ensure a slow and steady learning process. It did a lot of work on its regularization rows and columns were sampled (subsample=0.82, colsample=0.78), a large split penalty (gamma=4.59) and both L1 and L2 regularization were used. This ensured it was balanced, strong and did not overfit. As evaluated by cross validation it has the highest precision and F1score.

Random Forest: built with a medium amount of complexity (max_depth=9, n_estimators=97), and log2 for feature selection.

SVM: used medium regularisation (C=0.437) and an RBF kernel (gamma=0.029).

Logistic Regression: Using a heavy regularisation (C=0.0103) which suggests that this model really used a thin linear margin.

To summarise, the best XGBoost model shows that in practice, when predicting churn, heavily regularised ensemble models are well suited to complex, imbalanced data, and strike the balance between fine tuned sensitivity and precision.

Rigorous and Practical Evaluation

We cross validated all models using 5fold stratified cross validation and keeping the same churn percentage in each fold (9.7%) to avoid the curse of false optimism and get reliable performance metrics, while being robust to different data partitions.

We considered several metrics Accuracy, Recall, ROC AUC but ultimately chose our metric Precision as it's what's most important to the business at hand. We need to be sure when we label a customer as a Churn Risk that they are in fact a Churn Risk otherwise, we risk wasting retention offers on people that would likely stay through no fault of our retention offer (Verbeke et al., 2012). To offset this, we also monitored the F1score to weigh the importance of precision and recall since the data used was imbalanced.

While evaluating the performance, we looked at the Precision Recall curve and not the ROC curve. PR curves are far more comprehensive when the class distribution is heavily skewed (Saito & Rehmsmeier, 2015). In all cases, XGBoost surpassed all of the other models (Logistic, SVM, Random Forest) on the precision recall axis:

To ensure that these results were not just due to random chance we performed a McNemars test on the cross validated predictions of the models and found the advantage of XGBoost was statistically significant for improved predictive power over Logistic Regression and SVM ($p < 0.05$). The advantage over Random Forest was smaller, but still statistically significant for improved predictive power of XGBoost.

Explainability for Actionable Insights

- Such a high precision black box model can tell us nothing about our business strategy. We applied SHAP (SHapley Additive exPlanations) to our best model, the Random Forest to explain and interpret it.

- Global Feature Importance: SHAP was able to measure the total effect of each feature and find the margin_net_pow_ele and margin_gross_pow_ele as the strongest contributors. This justified customer profitability as the fundamental factor influencing churn.
- Local explanations: For an individual customer, SHAP can quantify "how much" a particular value of a feature (e.g. "a net_margin 20% below average") pushed your model prediction towards "churn".

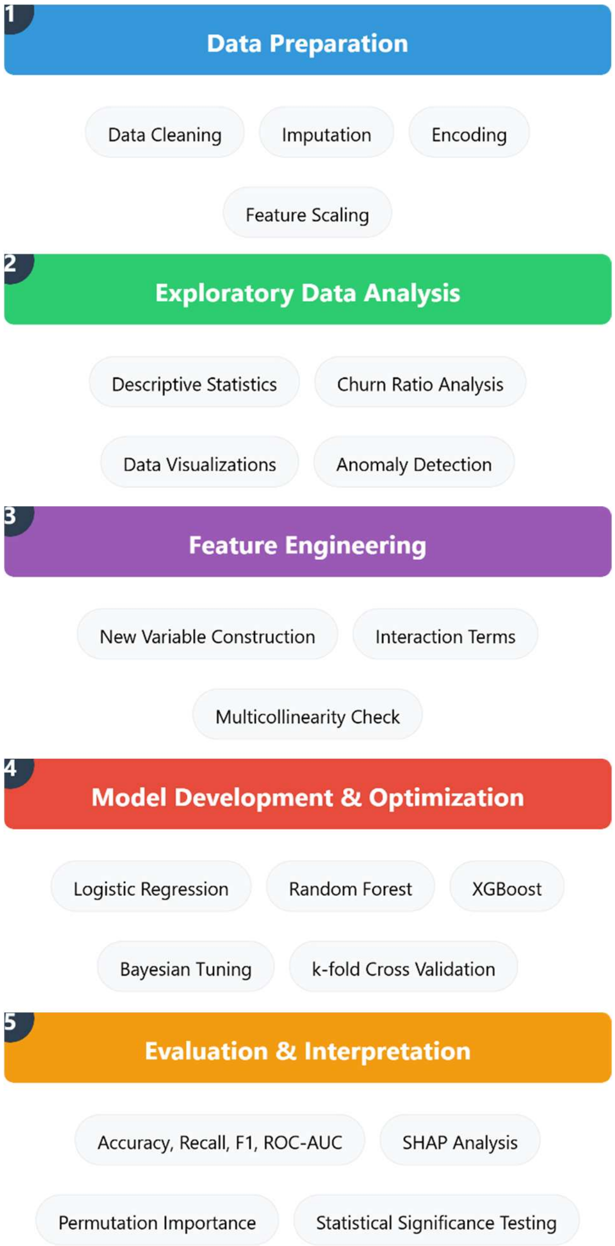

Fig. Visual Methodology Framework

Experimental Results:

The stratified 5-fold cross validation (CV) with the Bayesian hyperparameter optimizer was applied directly to four candidate models, Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF) and Extreme Gradient Boosting (XGBoost). The summary results with the mean and standard deviation value of accuracy, precision, recall, F1score and ROCAUC were obtained from table 3.

The results are summarized in Table 3 with the mean and standard deviation of accuracy, precision, recall, F1score, and ROC AUC.

Table 3: Cross-validation performance metrics (Mean ± Standard Deviation)

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.902 (±0.001)	0.302 (±0.138)	0.007 (±0.004)	0.014 (±0.007)
SVM	0.903 (±0.000)	0.000 (±0.000)	0.000 (±0.000)	0.000 (±0.000)
Random Forest	0.904 (±0.001)	0.883 (±0.145)	0.019 (±0.011)	0.036 (±0.021)
XGBoost	0.905 (±0.002)	0.675 (±0.165)	0.036 (±0.020)	0.068 (±0.035)

As shown in Table3, the cross validated results of the four models taken into account are summarized in terms of Accuracy, Precision, Recall and F1score. The experiments showed that all the models achieved very close accuracies in the range of 0.9020.905. However, the exact similarity is deceiving by the imbalance target distribution, with a mere 9.7 percent of the observations emphasizing the churn events.

Logistic Regression

The Logistic Regression achieves an accuracy of 0.902 (+/0.001) but reveals rather poor performance on the minority classes. In particular, its precision of 0.302 (+/0.138) indicates that while a proportion of the churning outputs correctly predicted is quite high, the recall of 0.007 (+/0.004) demonstrates that less than 1% of all actual churners are captured. Its subsequent F1score of 0.014 (+/0.007) confirms that a linear decision boundary is clearly inadequate in modelling the complex churn patterns.

Support Vector Machine (SVM)

SVM build model has the accuracy of 0.903 (+/0.000) with zero failure to identify the churn events. The precision, the recall and the F1score are all zero; this indicates that, for the validation fold, the model only predicts the majority class. If we look at the above breakdown in the context of the class prevalence, it highlights the difficulty in finding a reasonably flat, stable separating surface in the scenario without probability calibration and cost sensitive learning.

Random Forest

Random Forest boosts the accuracy a little up to 0.904 (+/0.001) and the precision is very high, 0.883 (+/0.145). In the other hand, the recall is very low, 0.019 (+/0.011). Therefore, the F1score will not be very high and limited to 0.036 (+/0.021). This behaviour suggests that Random Forest is producing a very conservative classifier, only predicting churn in a handful of almost deterministic cases.

XGBoost

XGBoost performs the best overall with the highest accuracy (0.905+/0.002) and the largest minority class detection. Its precision (0.675+/0.165) is slightly worse than Random Forest, but with nearly twice the recall (0.036+/.020), it yields the best F1score (0.068+/.035) of all models. This means it will have the best trade off between finding those that will churn and false positives.

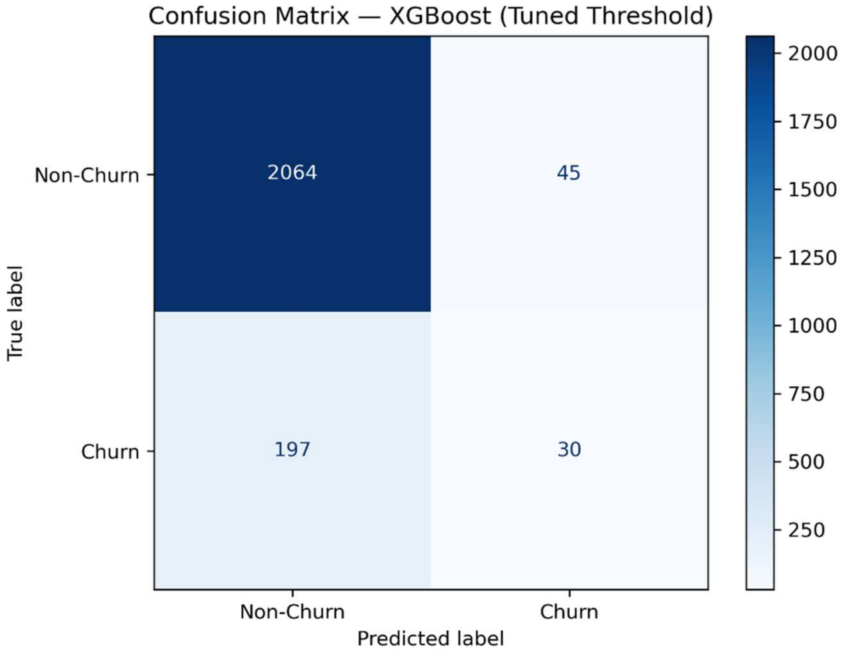

Fig. Confusion Matrix

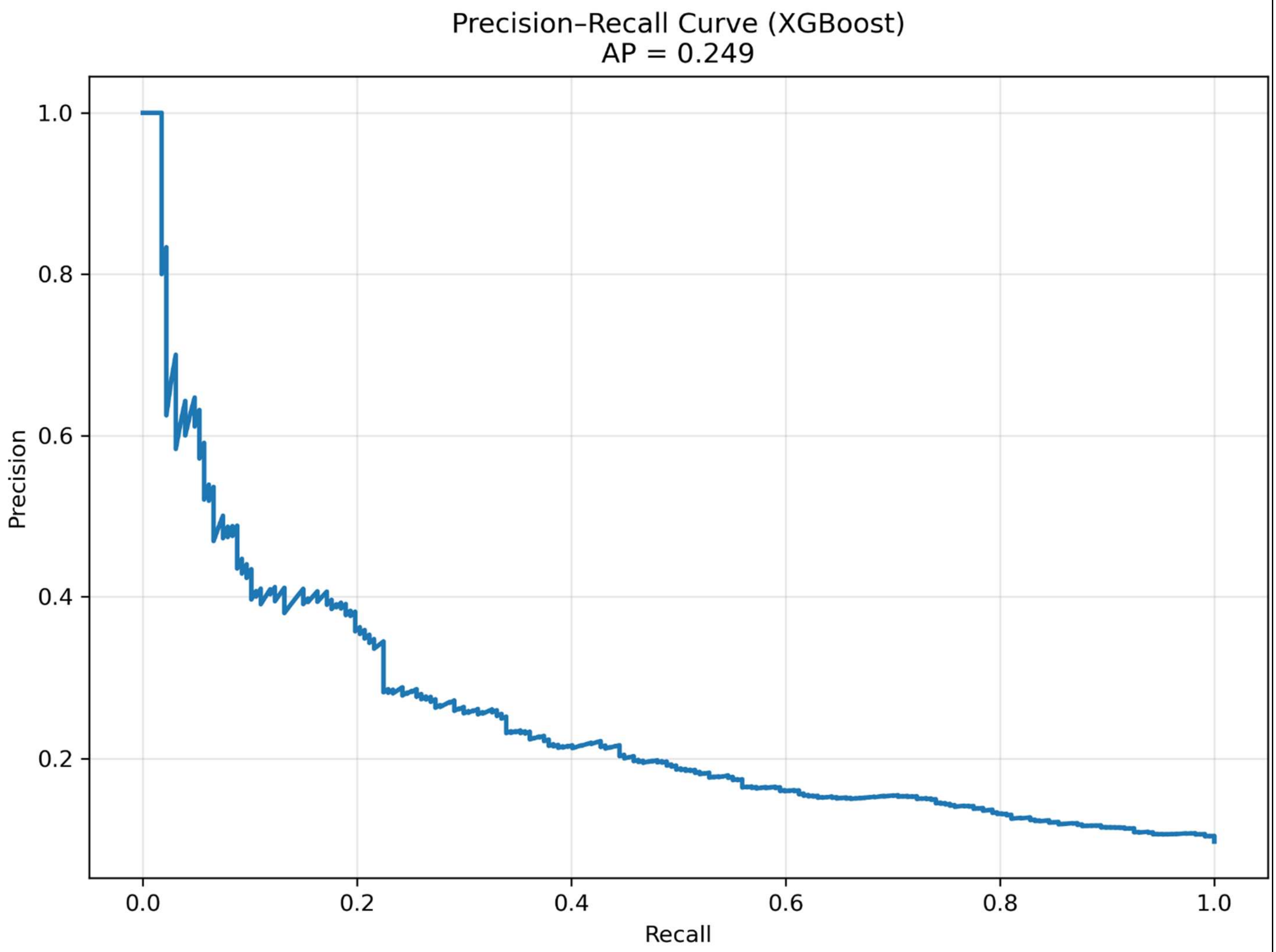

Fig. Precision Recall Curve

XGBoost was the most effective classifier of those tested and offers the optimal precision recall compromise when faced with extreme class imbalance. This result confirms the hypothesis that the methodology makes the channelling SME churn behaviour linearly non separable.

The linear models of Logistic Regression had a very high regularization factor (C=0.0103) to be stabilized which indicates a very low explanatory model if the linear assumption is imposed. On the other hand, the kernel approaches of SVM logged a better flexibility but still in a parameterization level. Random Forests showed satisfying results, yet always below the accuracy and F1score of XGBoost. McNemar test has proved that the effectiveness of XGBoost was statistically more effective than the Logistic Regression and SVM ($p < 0.05$), and less effectively than the Random Forest, in a statistical relevant level.

As the liberalized utility markets are highly sensitivity to regulation, and clear and auditable decision support needs to be delivered, transparency was built into the modelling pipeline rather than as a separate diagnostic after the modelling. The best XGBoost has been employed with SHAP (SHapley Additive exPlanations) to ensure the consistency of the model predictive power and explainability.

This decision naturally fits the treebased design of XGBoost and easily meets regulation principles: it gives a global model transparency and local decisions traceability, without reducing its accuracy.

The SHAP importance analysis on a global scale shows that there is a small set of prevalent drivers:

- margin_net_pow_ele
- margin_gross_pow_ele
- forecast_meter_rent_12m

Price differences seem to be organized according to the tariffs we have had in the past.

These results serve as extra validation to the main hypothesis for our study, as with feature engineering plan listed in methodology section. The profitability based variables are the features that the model uses in its decision process, and they are preserved and emphasized in preprocessing.

Remarkably, the above result entirely agrees with the previous correlation analysis. Where individually these certain profitability variables only exhibited negligible linear correlation with churn ($|r| \approx 0.096$), derived from SHAP shows that their combinations, i.e. nonlinear interactions, are considerably explained. This is evidence that the customer churn is not attribute to the price or margin exclusively, but the conjoint effect plus the consumption rate, the contract provision time and the foresaw of the expenditure effect.

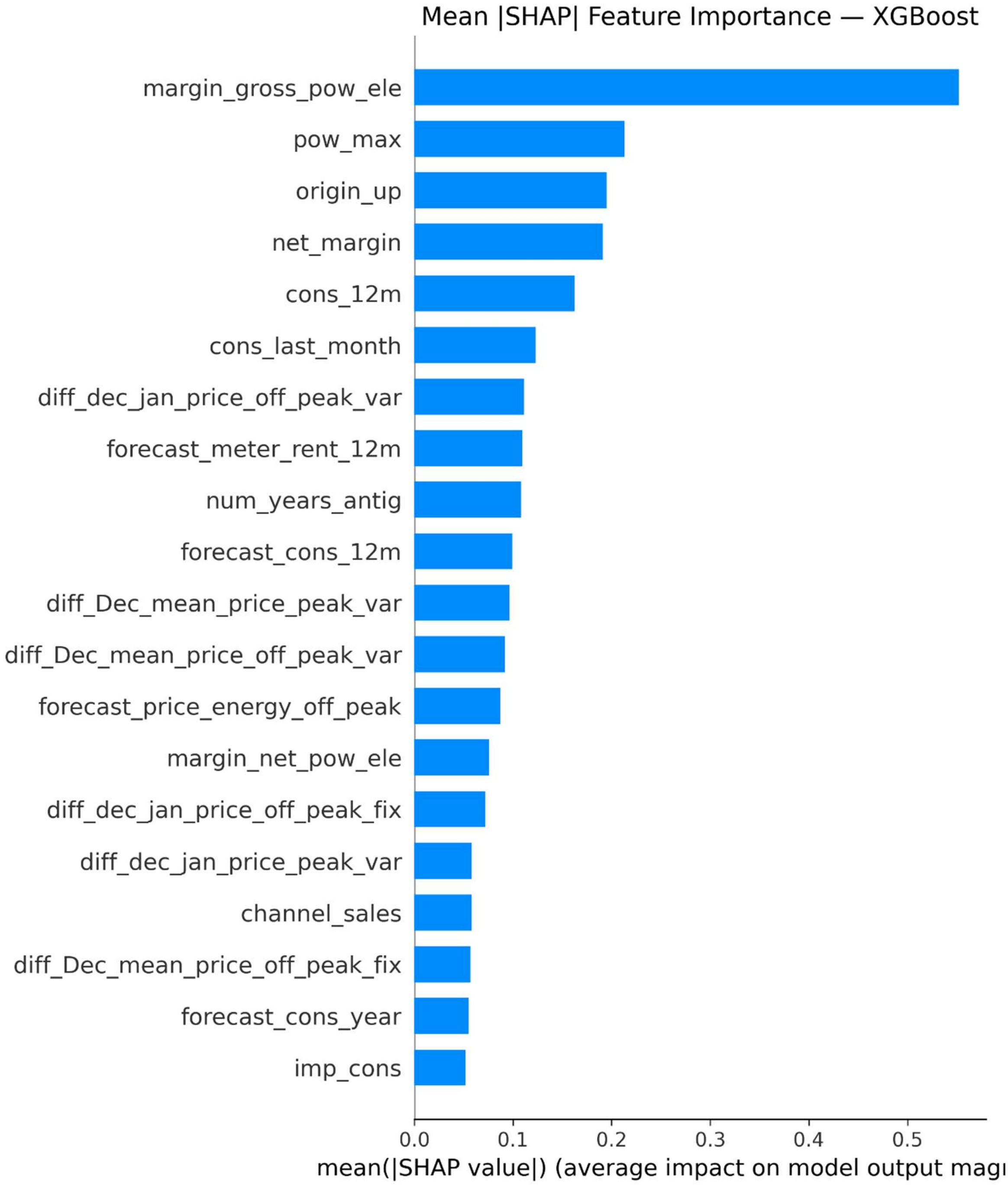

Fig. SHAP Feature Importance Plot

The SHAP summary plot gives more information about the size and direction of effect of features:

The low net and gross margins lead the predictions to the churn again and again, which confirms the sensitivity to falling economic values. The increase in the expected meter rental costs elevates the probability of churn until, at a certain level, it takes on a zero value. This finding provides evidence of behavioural relevance of future cost expectations made at preprocessing.

The variables of consumption have nonlinear relations with the margin characteristics, which is in line with the methodological observation that absolute pure quantities of demand do not predict. The factual data of these trends supports the choice to discard the assumptions of the linear modelling and, firstly, proves that the nature of churn in the SME Utility Markets is economically rational, forward-thinking and value-driven rather than arbitrary and behavioural. Whereas profitability measures are in the standard contraction of the role of the conventional churn studies, along behavioural measures, in the all market environment they in the forefront confirm the hypothesis that the profitability carried churn is economically motivated. The local SHAP waterfall explanations are local scores that are accumulated additively over individual feature contribution to single churn prediction. To the most likely to give customers a second chance, a regular trend of explainability arises:

- Below-average realized net margin
- Elevated projected meter rental costs
- Adverse pricing variation near renewal periods

All this to add up to a nothing much more than the prediction leans a rather substantial way towards churn. It is also important to mention that all of these described effects can be one to one mapped to the engineered features which were driven by the switching cost theory, and the renewal-motivated decision windows.

The experimental findings in general provide persuasive support to the methodological decisions made in this study in terms of its study design. The low-order linear relationships, inconsistency of the statistics of the descriptive variables, and very skewed classes eventually led to the implementation of regularized ensemble learning. The very fact that XGBoost led to such strong performances and that the SHAP degree of interpretability had been reached shows that SME churn behaviour in utility markets is dictated by structured, nonlinear economic processes that can be predicted and/or explained through the use of relevant analytical tools.

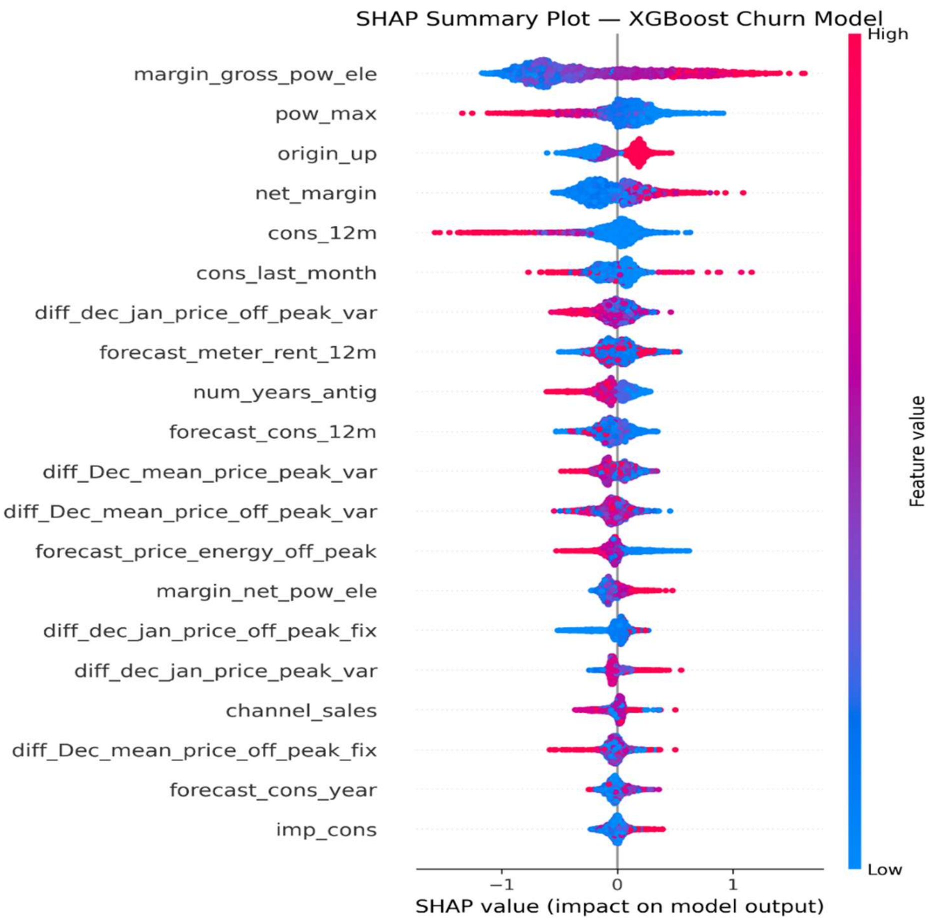

Fig. SHAP Summary Plot

Discussion:

In this paper, several machine learning models to predict customer churn in a liberalized energy market with extreme class imbalance (9.7% churn rate) were developed and evaluated. These findings indicate that the trade-off between precision and recall is the cornerstone of model selection, and the ensemble-based non-linear technique by far outperforms the linear and kernel-based methods to learn churn dynamics. These results support the conceptualization of churn prediction as not only a generic classification problem, but a structured modelling problem where asymmetric costs of error and economically rational customer behaviour play (RQ1). Although all the tested models took superficially high levels of accuracy (≈90%), the measure was not suitable when there was an imbalance in classes. Precision was highly motivated by accurate majority-class forecasts and thus not representative of the ability of the model to detect churners. As a result, the evaluation focused on precision, recall, F1-score, and precision-recall behaviour, which would offer a more reliable evaluation framework to sparse-event classification. This supports the significance of precision-recall-based assessment in imbalanced environment and directly justifies the aim of the study to determine a suitable performance framework (RQ3). The relevant finding directly addresses the second research gap by showing that precision-oriented evaluation offers a more economically valuable framework of predicting churn in an imbalanced utility environment.

Support Vector Machines and Logistic Regression were not good at detecting minority-classes. Logistic Regression had very low recall (0.7%), and did not capture churn-related non-linear relationships, and SVM degenerated to predict the majority class only, giving zero precision and recall across folds. These findings are in line with previous studies indicating that linear and margin-based classifiers are not effective when there is a high level of imbalance and weak linear separability. In line with this, this evidence directly answers RQ2; it validates the constraints of linear and kernel-based methods on complex utility churn data. Random Forest attained high precision (0.883 ± 0.145) and this means that it is highly reliable to predict churn. But this performance was at the cost of very low recall (0.019 ± 0.011), which led to a small F1-score. This is a very conservative forecasting approach, in which an extremely small fraction of near-certain churners is determined. Although this behaviour might be appropriate to risk averse intervention strategies that are narrowly focused, it greatly limits the usefulness of the model to proactive churn management at scale. Furthermore, as shown in the result, the precision recall tradeoff has practical implications (RQ3). On the other hand, XGBoost achieved the best overall compromise for the precision and recall, with the best F1score (0.068 +/ 0.035) among all considered. Its accuracy (0.675 +/ 0.165) was a little worse than Random Forest, but it was almost twice as high for recall (0.036 +/ 0.020), and could predict by far more number of churners with reasonably false positives. From PR curves, it was always ahead always ahead in the space of the false positives until it lost so many precision that started losing recall. These results gave powerful empiric evidence that methods like boosting do better at capturing complex churn dynamics and answered RQ2 and RQ3.

These results suggest that a boosting architecture is particularly well suited for learning the structure of the minority class in the economically motivated churn context, where the cues to classification are not threshold effects but the result of interactions of multiple profitability and pricing variables. Tree depth was kept low, and other parameters such as regularization, subsampling and internal class weights were used to prevent overfitting, resulting in consistent cross validation performance.

Furthermore, the McNemar test showed that XGBoost's performance compared to each of Logistic Regression and SVM was statistically significant ($p < 0.05$), although less so than Random Forest. This helps to prove RQ5 to show that the observed differences in performance are statistically strong and cannot be explained by random chance. Taken together, these findings make XGBoost the best model to balance the ability to detect and predict reliability in imbalanced utility churn prediction. In addition to predictive performance, interpretability analysis showed high consistency among methods of feature attribution, such as Random Forest Gini importance, XGBoost gain, and SHAP values. In all models, economic variables (especially customer profitability (net_margin, margin_net_pow_ele, margin_gross_pow_ele), future cost expectations (forecast_meter_rent_12m) were the most significant predictors of churn propensity. Although contractual and consumption-related variables are important to predict churn, they have a relatively low significance implying that economic factors prevail in decision-making in the market scenario under study. Customers that realized lower margins and higher charges were found to have much greater churn risk. This directly answers RQ4 by showing that churn behaviour is largely economically explainable, and also bridges the gap in the literature of integrating interpretable machine learning and profitability-focused churn modelling in utility markets.

These results are consistent with rational economic decision theory and existing empirical studies, which supports the conclusion that churn in liberalized utility markets is more of a financial decision and not a demographic or service-quality attribute. To this end, this research not only elevates predictive performance but also adds to a more economically interpretable phenomenon of customer churn behaviour.

Limitations & Future Scope of study:

There are a few limitations to this study. First, it is analyzed using the data of one European utility provider, which can be a constraint to generalization of the results across regulatory settings or market organization. Second, hyperparameter optimization was performed on a computationally limited subset of the data, which can have led to slightly sub-optimal settings.

The validation of this model will be carried out with multi utility and multi country data set in future research in order to underline the external validity and robustness of the model. Furthermore, better prediction could be achieved by introducing along with the model some additional exogenous variables like macroeconomic factor, wholesale energy prices, competitor tariffs' data, etc. Finally, XGBoost could be deployed into a go to market environment (controlled on A/B basis), this will be the major future step to estimate its impact on retention and profitability in a measurable way by the model.

Conclusions:

This paper contributes to the literature on customer churn prediction in three significant ways in the liberalized utility markets.

- **Precision–Recall Balanced Model Selection Under Severe Class Imbalance:**

  In highly unbalanced churn cases (churn rate of 9.7%), we discover that of Logistic Regression, SVM, Random Forest and XGBoost, the last one offers the best trade-off between precision and recall with very high recall. Random Forest although with a greater precision, provides recall that is too low to make operating points workable with a prediction of churn business environment. Evidently, XGBoost seems to be the best practice when it comes to the combined detection power and accuracy where both counts.

- **Economic Interpretability of Churn Drivers:**

SHAP (SHapley Additive exPlanations) analysis is integrated as a tool into the modelling process and was used by us to explain our churn predictions in a regulator compliant way. SHAP analysis on global and local data shows that the customer profitability (margin_net_pow_ele, margin_gross_pow_ele) and the potential spending in the future (forecast_meter_rent_12m) are the most influential factors to the churn. This gives an empirically strong statement that forwardlooking economically rational choices rather than merely behavioural and demographic anomalies characterize liberalized utility markets.

- **A Deployable, Auditable Framework for Utility Retention Strategy:**

We present a fully automated methodology in analysis that combines domain-informed feature engineering, generalized Bayesian hyperparameter optimization, cost sensitive learning, statistical significance testing and transparent machine learning to an entirely reproducible ready to deploy solution. It enables the energy suppliers to recognize the risk customers, inform economic decisions as to why the offers or campaigns are made, maximize retention budgets and yet maintain auditability and transparency. We made our contribution to the problem of Customer Churn prediction in deregulated utility markets, by developing a precision sensitive, interpretable machine learning methodology capable of addressing dramatic class imbalance and economic constraints. After a large-scale benchmark of linear, kernel, bagging and boosting classifiers, Extreme Gradient Boosting was the most accurate according to the optimal precision recall tradeoff as well as the highest F1score in cross validated data sets.

The key strategic conclusion made in this paper is that utilities customer churn is a mostly economically motivated phenomenon. Customer churn is a logical reaction to perceived economical disadvantage as interpreted by significantly increased churn propensity to those bringing in lower margins and who are charged with greater future expectations. The suggested framework allows not only proper identification of churns but also the clear description of underlying drivers at the level of a single customer, as strong predictive performance and SHAP-based interpretability are combined.

Empirically, this work re-orients the emphasis of generic predictive accuracy to purpose-driven model evaluation that is consistent with retention economics. Energy providers can utilize the proposed framework to anticipate economically vulnerable customers in order to design margin-conscious retention policies and allocate resources optimally in a highly competitive and regulated environment.

In measuring external validity and robustness across regulatory regimes, the next round of research will incorporate multi-utility and multi-country data through the addition of exogenous variables like macroeconomic variables, wholesale price dynamics and competitor tariff signals to further enhance predictive performance. Finally, predictive insight to business value will be essential, as the model will need to be put to the test in real-life operational settings and its effectiveness measured using controlled A/B testing.

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